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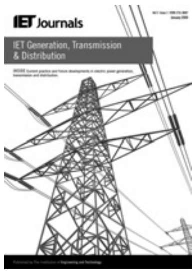

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Instantaneous positive-sequence current applied for detecting voltage sag sources

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Abstract: This study proposes a method for detecting voltage-sag sources, using only the measurements of line currents. Instantaneous symmetrical components are introduced for accommodating the transient nature of voltage sag events, where a magnitude and phase change in the instantaneous positive-sequence current is used as a criterion. The proposed method is compared with conventional phasor-based methods known from the literature. All the discussed methods for voltage sag source detection, both the already known and the proposed ones, were evaluated by applying extensive simulations, laboratory tests and field testing. When compared with the phasor-based methods, the proposed method showed faster response, a higher degree of confidence and superior effectiveness, especially for transient voltage sag events in networks with distributed generation and active loads.

1 Introduction

Voltage sags (dips) are common transient disturbance events [1, 2]. As they can be provoked by different events related to a transmission or distribution network or even to the customer, the frequency of voltage sags can be between a few ten and even several thousand times per year [3]. The impact of disturbances caused by voltage sags on production losses, different types of electrical machines and industrial equipment have already been reported [4–11].

In those cases when the mitigation equipment is not installed, any disputes can only be resolved fairly if the voltage sag source is reliably detected. Several methods for the detecting and tracking of voltage sag sources have already been reported and reviewed [12–44]. Methods for upstream–downstream detections of voltage sag sources are based on different criteria, such as disturbance energy [12–16], voltage-current characteristics [17, 18], incremental impedance [19–22], voltage characteristics [27, 28] and current change [29, 30]. Furthermore, statistical methods have been applied to increase the effectiveness and degree of confidence. Methods with combined rules have been developed in this way however, they are all based on stationary phasors [25, 26, 36]. Higher effectiveness has also been achieved for several types of methods by applying instantaneous values [23, 24].

The majority of the described methods [12–26] require measurements of both, line voltages and currents, which is sometimes unavailable. In distribution networks, directional overcurrent relays may be applied for detecting voltage sag sources [29, 30], where magnitude and phase changes within the positive-sequence current phasor are used as a criterion. It is because voltage sags are transient disturbance events, conventional phasor-based methods might produce questionable results. In order to overcome this difficulty, a method is introduced for the detections of voltage sag sources, where the criterion is based on magnitude and phase changes in the instantaneous positive-sequence current [45]. All the discussed methods, both the already known and the proposed ones, can be used for detecting voltage sag sources caused by different events related to the network, such as faults, heavy motor starting or transformer energising. However, these methods cannot distinguish between fault and non-fault voltage sag events.

All the discussed methods that use only current measurements were tested for different types of voltage sags in a radial distribution network with distributed generation (DG) and active loads. In order to evaluate the discussed methods, extensive numerical simulations, laboratory testing and field testing were performed. On the basis of the obtained results, the correctness of the proposed method is evaluated.

2 Detection of voltage sag sources using only the measurements of line currents

Voltage sags are because of short-duration increases in currents. Therefore, currents measured at the monitoring point (MP) increase during downstream events and decrease during upstream events (Fig. 1). Two methods [29, 30] have already been proposed based on this assumption, which are both based on positive-sequence current phasor for the fundamental frequency component. Conventional phasor estimation techniques, such as discrete Fourier transform [46] or signal least-square (LSQ) [47] might produce inconclusive results because of the inherent averaging of the input signals. Kalman filtering [48] was

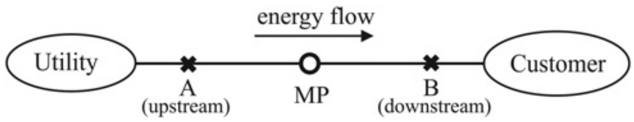

Fig. 1 *Upstream (A) and downstream (B) voltage sag event*

therefore applied within the method proposed in [29]. However, parameterisation of such filters requires *a-priori* information about the signals' harmonic contents and noise properties, which is a serious obstacle in practice. Therefore instantaneous symmetrical components are introduced, where only instantaneous positive-sequence component is taken into account.

2.1 Instantaneous and steady-state symmetrical components

The instantaneous line currents i_k [For simplicity, i is used instead of $i(t)$ for denoting instantaneous values.], where $k \in \{a, b, c\}$ (a, b and c denote individual phases), may be transformed by

$$\begin{bmatrix} \underline{i}_+ \\ \underline{i}_- \\ i_0 \end{bmatrix} = \underline{\boldsymbol{F}} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \tag{1}$$

$$\underline{\boldsymbol{F}} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & \underline{a} & \underline{a}^2 \\ 1 & \underline{a}^2 & \underline{a} \\ 1 & 1 & 1 \end{bmatrix}, \quad \underline{\boldsymbol{F}}^{-1} = (\underline{\boldsymbol{F}}^*)^{\mathrm{T}} \tag{2}$$

The unitary transformation matrix (2), where $\underline{a} = \mathrm{e}^{\mathrm{j}2\pi/3}$ and $(\cdot)^*$ denotes the complex conjugate, follows the decomposition proposed by Fortescue [45]. Thus, the transformed currents and voltages form the instantaneous positive, negative and zero-sequence components. The instantaneous positive and negative-sequence components are complex conjugates, that is, $\underline{i}_+ = \underline{i}_-^*$, whereas the instantaneous zero-sequence component i_0 is always real.

Steady-state line currents are sinusoidal quantities with the fundamental angular frequency ω. Thus, they can be represented as the sum of two complex conjugated terms

$$i_k = \frac{\sqrt{2}}{2}\left(\underline{I}_k\,\mathrm{e}^{\mathrm{j}\omega t} + \underline{I}_k^*\,\mathrm{e}^{-\mathrm{j}\omega t}\right) \tag{3}$$

where $\underline{I}_k$ are the current phasors. Transformation of steady-state line currents (3) using the unitary matrix (2) yields to definition of phasors for the positive, negative and zero-sequence components

$$\begin{bmatrix} \underline{I}_+ \\ \underline{I}_- \\ \underline{I}_0 \end{bmatrix} = \underline{\boldsymbol{F}} \begin{bmatrix} \underline{I}_a \\ \underline{I}_b \\ \underline{I}_c \end{bmatrix} \tag{4}$$

2.2 Existing methods based on a positive-sequence current phasor

Two methods, which use only measurements of line currents have already been proposed [29, 30]. The first method [29] is based on a phase change within the positive-sequence current phasor. It has been developed for fault direction estimation in radial distribution networks with DG. The second method [30] uses magnitude change in the positive-sequence current phasor as an additional criterion. Note, that only the extended method [30] is discussed further on and is given by

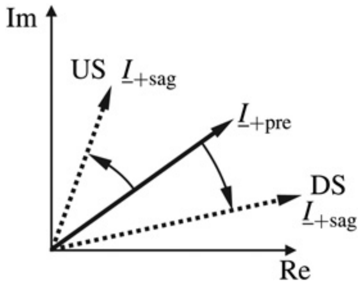

Fig. 2 *Graphic presentation of change in the positive-sequence current phasor for upstream (US) and downstream (DS) voltage sag events*

$$\begin{array}{llll} \text{if } \Delta\angle\underline{I}_+ > 0 & \text{or} & \Delta|\underline{I}_+| < 0 & \Rightarrow \text{US} \\ \text{elseif } \Delta\angle\underline{I}_+ < 0 & \text{and} & \Delta|\underline{I}_+| \geq 0 & \Rightarrow \text{DS} \end{array} \tag{5}$$

where 'US' and 'DS' denote upstream and downstream voltage sag events, respectively. The magnitude change within the positive-sequence current $\Delta|\underline{I}_+|$ is defined by the difference $|\underline{I}_{+\text{sag}}| - |\underline{I}_{+\text{pre}}|$, whereas the phase change $\Delta\angle\underline{I}_+$ is defined by the difference $\angle\underline{I}_{+\text{sag}} - \angle\underline{I}_{+\text{pre}}$ within the interval $[-\pi, +\pi]$. A pre-sag and during-sag state are denoted by the subscripts 'pre' and 'sag, respectively (Fig. 2).

2.3 Magnitude and phase change within an instantaneous positive-sequence current

The magnitude and phase of the instantaneous positive-sequence current $\underline{i}_+$ depend on its real and imaginary components $\mathrm{Re}\{\underline{i}_+\}$ and $\mathrm{Im}\{\underline{i}_+\}$, which are time dependent functions. In the pre-sag state they are normally sinusoidal waveforms shifted by $\pi/2$, while their magnitudes are equal (Fig. 3*a*). Calculations of the magnitude and phase change are performed over the following steps:

Step 1: Within the observed time window line currents are transformed by (1), (2). The obtained time dependent functions are $\mathrm{Re}\{\underline{i}_{+\text{sag}}\}$ and $\mathrm{Im}\{\underline{i}_{+\text{sag}}\}$ (black lines in Fig. 3*a*).
Step 2: The last cycle of the obtained time dependent functions prior to the sag inception is extended to obtain $\mathrm{Re}\{\underline{i}_{+\text{pre}}\}$ and $\mathrm{Im}\{\underline{i}_{+\text{pre}}\}$ (grey lines in Fig. 3*a*).
Step 3: Magnitude and phase are calculated individually for $\underline{i}_{+\text{pre}}$ and $\underline{i}_{+\text{sag}}$.
Step 4: Magnitude change $\Delta|\underline{i}_+|$ is calculated by $|\underline{i}_{+\text{sag}}| - |\underline{i}_{+\text{pre}}|$, whereas phase change $\Delta\angle\underline{i}_+$ is calculated by $\angle\underline{i}_{+\text{sag}} - \angle\underline{i}_{+\text{pre}}$ within the interval $[-\pi, +\pi]$.

An example is shown in Fig. 3 for a downstream voltage sag event because of a phase-to-phase (P-P) fault. Functions $\mathrm{Re}\{\underline{i}_+\}$ and $\mathrm{Im}\{\underline{i}_+\}$ both changed in the magnitude as well as in the phase (black lines in Fig. 3*a*), with reference to the expected normal sinusoidal waveforms (grey lines in Fig. 3*a*). The obtained magnitude change $\Delta|\underline{i}_+|$ is positive for this case (Fig. 3*c*), whereas the resulting phase change $\Delta\angle\underline{i}_+$ is negative (Fig. 3*d*) because of the lag in the phase $\angle\underline{i}_{+\text{sag}}$ (Fig. 3*b*). According to the generalisation of the

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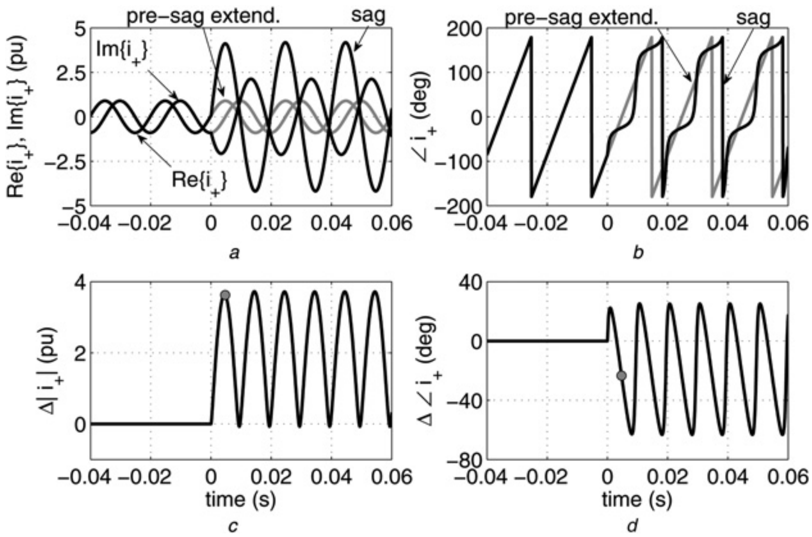

Fig. 3 *Instantaneous positive-sequence current for downstream P-P fault – numerical simulations*

a Real and imaginary components
b Phase
c Magnitude change
d Phase change

method (5) the obtained results correctly indicate a downstream event for this case.

2.4 Generalisation of the phasor-based method

The conventional phasor-based method (5) may be generalised by replacing the positive-sequence current phasor $\underline{I}_+$ with the instantaneous positive-sequence current $\underline{i}_+$. In order to obtain conclusive results the disturbing influence of the alternating component in time responses of $\Delta|\underline{i}_+|$ and $\Delta\angle\underline{i}_+$ (Figs. 3*c* and *d*) should be taken into account. A similar principle can be used as within the energy-based method [12, 24], where the direction of the voltage sag source is detected by the sign of the instantaneous disturbance energy at the end of the event, defined by $\Delta w = \int_0^t (p_{\text{sag}} - p_{\text{pre}})\,\mathrm{d}\tau$. Thus, the proposed current-based method is given by the integral criteria

$$\begin{array}{llll} \text{if } \int_0^t \Delta\angle\underline{i}_+\,\mathrm{d}\tau > 0 & \text{or} & \int_0^t \Delta|\underline{i}_+|\,\mathrm{d}\tau < 0 & \Rightarrow \text{US} \\ \text{elseif } \int_0^t \Delta\angle\underline{i}_+\,\mathrm{d}\tau < 0 & \text{and} & \int_0^t \Delta|\underline{i}_+|\,\mathrm{d}\tau \geq 0 & \Rightarrow \text{DS} \end{array} \quad (6)$$

To increase the degree of confidence additional criteria are proposed, similar to that in [12], where the sign of the initial peak in the instantaneous disturbance power $\Delta p := p_{\text{sag}} - p_{\text{pre}}$ is taken into account. As the initial peak in the magnitude change $\Delta|\underline{i}_+|$ and the initial peak in the disturbance power Δp are typically only slightly shifted (circular marks in Figs. 3*c* and 4*b*), the additional criteria

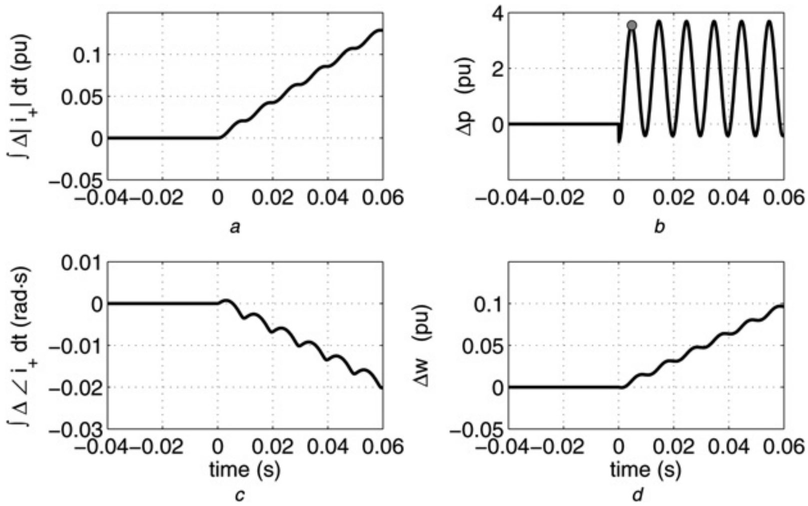

Fig. 4 *Integral and the disturbance energy criteria for a downstream P-P fault (numerical simulations – Fig. 3)*

a, *c* Magnitude and phase changes within instantaneous positive-sequence current
b Disturbance power
d Disturbance energy

are given by

$$\begin{array}{llll} \text{if } \Delta\angle\, \underline{i}_+(t_p) > 0 & \text{or} & \Delta|\underline{i}_+|(t_p) < 0 & \Rightarrow \text{US} \\ \text{elseif } \Delta\angle\, \underline{i}_+(t_p) < 0 & \text{and} & \Delta|\underline{i}_+|(t_p) \geq 0 & \Rightarrow \text{DS} \end{array} \quad (7)$$

where t_p is the time instant after the sag inception when the initial peak is reached within the magnitude change (circular mark in Fig. 3*c*). When the results obtained by (6) and (7) are equal, then the degree of confidence is high. However, when the results are different, the voltage sag event is in the direction indicated by the integral criteria (6), but with a lower degree of confidence.

The results obtained for the discussed example indicate a downstream event with a high degree of confidence because the signs of $\Delta|\underline{i}_+|$ and $\Delta\angle\, \underline{i}_+$ (circular marks in Figs. 3*c* and *d*) match the signs of their integrals at the end of the event (Figs. 4*a* and *c*). Furthermore, the same result, that is, downstream, is determined using the energy-based method, as Δp and Δw are both positive (Figs. 4*b* and *d*).

3 Results

3.1 Influences of transient events

To evaluate the influences of transient events, a radial distribution network was considered, as is shown in Fig. 5. A balanced passive inductive load and a DG source were considered on the customer side. In order to provoke the voltage sag event, a P-P fault was considered at the transformer's primary side. A transient response was expected because of the exhaustion of energy generated by the DG source during the event. Note that an induction generator was used to simulate the DG source.

The downstream voltage sag was considered at the DG source terminals (MP1). The results obtained using the phasor-based method show a typical time delay of approximately one cycle (Figs. 6*a* and *b*). The phase change is indeed negative (Fig. 6*b*), but the sign of the magnitude change varies over time (Fig. 6*a*). Consequently, the obtained results were inconclusive. The results obtained using the proposed method showed a positive value of the integral for the magnitude change, whereas the value of the integral for the phase change was negative (Figs. 6*e* and *f*), which correctly indicated a downstream event. Furthermore, the obtained results showed the highest degree of confidence, as the signs of the magnitude and phase changes at time instant t_p (circular marks in Figs. 6*c* and *d*) were the same as the signs of their integrals at the end of the event (Figs. 6*e* and *f*).

When the currents were captured at the terminals of a passive inductive load (MP2) an upstream voltage sag was considered. In order to detect upstream events only one criterion should be fulfilled, that is, negative change in the magnitude or positive change in the phase. The first peak of the instantaneous magnitude change and its integral at the end of the event were both negative (Figs. 7*c* and *e*). Thus, the proposed method correctly indicated an upstream event. Furthermore, the obtained results did not show the highest degree of confidence, as the value of the phase change at time instant t_p was significantly close to zero (circular mark in Fig. 7*d*), while its integral was negative (Fig. 7*f*). Results obtained using the phasor-based method might be correct as the magnitude change was negative (Fig. 7*a*). However, the phase change should always be negative because the monitoring point (MP2) was located at the terminals of a passive inductive load. The sign of the phase change varied over time (Fig. 7*b*), thus the obtained results were considered to be inconclusive.

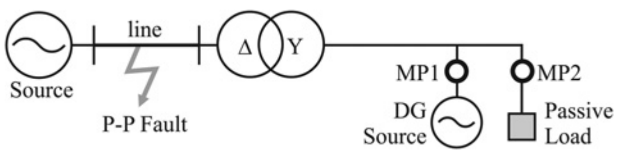

Fig. 5 *Radial distribution network with DG during a voltage sag event*

3.2 Laboratory and field testing

To confirm the proposed method the same experimental laboratory set-up was used as in [24]. An induction machine was connected to the network through a Dyg5 power transformer. Line currents were captured by a dSPACE DS1103 PPC digital control system at a single MP where a sampling frequency of 10 kHz was used. Different types of faults were generated at the transformer's primary and secondary sides, as well as at the machine terminals: phase-to-earth (P-E), phase-to-phase-to-earth (P-P-E), P-P and three-phase (3-P). Tests were performed separately for the motor and generator operations of the induction machine. Altogether 22 examples of voltage sags were generated in this way, half of them in the upstream and half of them in the downstream directions. The proposed method failed only within two downstream events because of P-E and P-P-E faults for a system with a generator. On the other hand, the phasor-based method failed within all downstream events, where the results were incorrect for a system with a generator and inconclusive for a network with a motor. An example is shown in Fig. 8, where a downstream sag event was considered because of the P-P fault at the motor terminals. Magnitude change was positive within both methods (Figs. 8*a*, *c* and *e*), which is correct. The proposed method indicates a downstream event with a high degree of confidence, as the phase change at time instant t_p and its integral at the end of the event were both negative (Figs. 8*d* and *f*). The result obtained by the phasor-based method was inconclusive, as the sign of the obtained phase change varied rapidly over time (Fig. 8*b*).

Field tests were also applied to the discussed methods. Currents were captured in the Slovenian distribution network at the 20 kV level by protective relays at the utility and at the customer side. Only downstream voltage sags were analysed, altogether 11 examples (4 because of P-E faults, 2 because of P-P faults and 5 because of 3-P faults). The proposed method failed only twice for P-E faults, whereas the phasor-based method failed for both P-P faults as well as for an additional P-E fault (altogether five examples). Fig. 9 shows results obtained for a downstream P-E fault where both methods failed as the sign of the phase change was incorrect (Figs. 9*b*, *d* and *f*) because of the disturbing influence of the high resistance earthing. Voltage sags because of the transformer energising were also captured during the field testing. Two examples were analysed for the downstream direction at the 20 and 110 kV levels, and one example for the upstream direction at the 0.4 kV level. Both methods provided correct results for the downstream sags, whereas for the upstream sag only the proposed method was successful.

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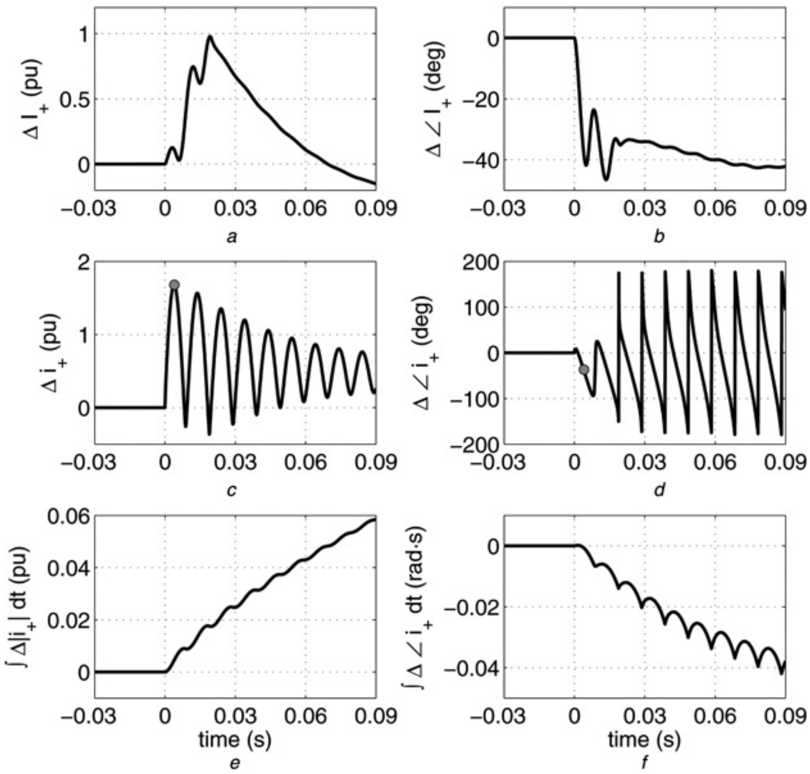

Fig. 6 *Numerical simulations for a downstream voltage sag (MP1 in Fig. 5)*

a, *b* Magnitude and phase change (inconclusive results because of the magnitude change sign variation)
c, *d* Instantaneous magnitude and phase change
e, *f* Integrals for instantaneous magnitude and phase change

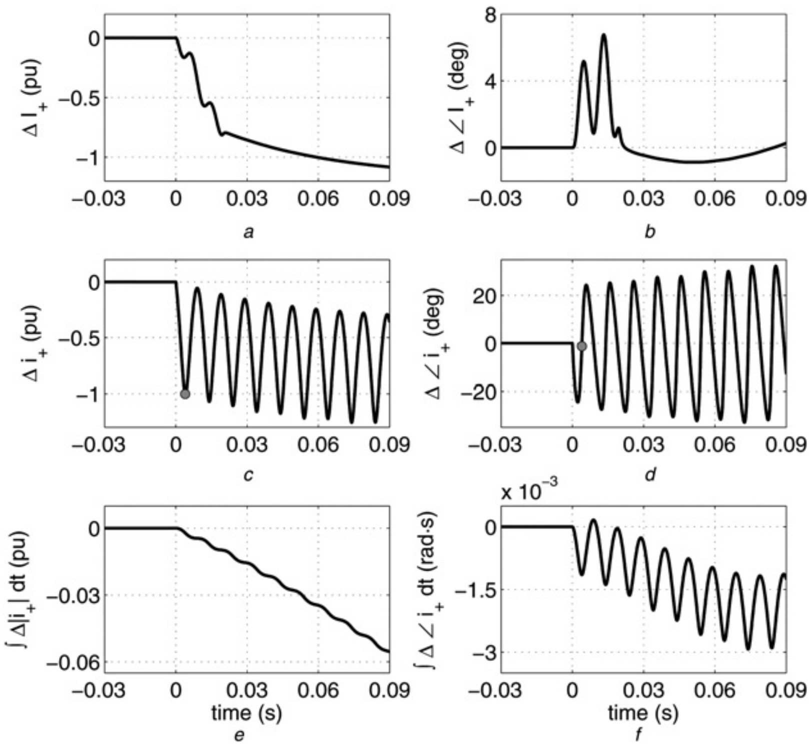

Fig. 7 *Numerical simulations for an upstream voltage sag (MP2 in Fig. 5)*

a, *b* Magnitude and phase change (inconclusive results because of the phase change sign variation)
c, *d* Instantaneous magnitude and phase change e, f Integrals for instantaneous magnitude and phase change

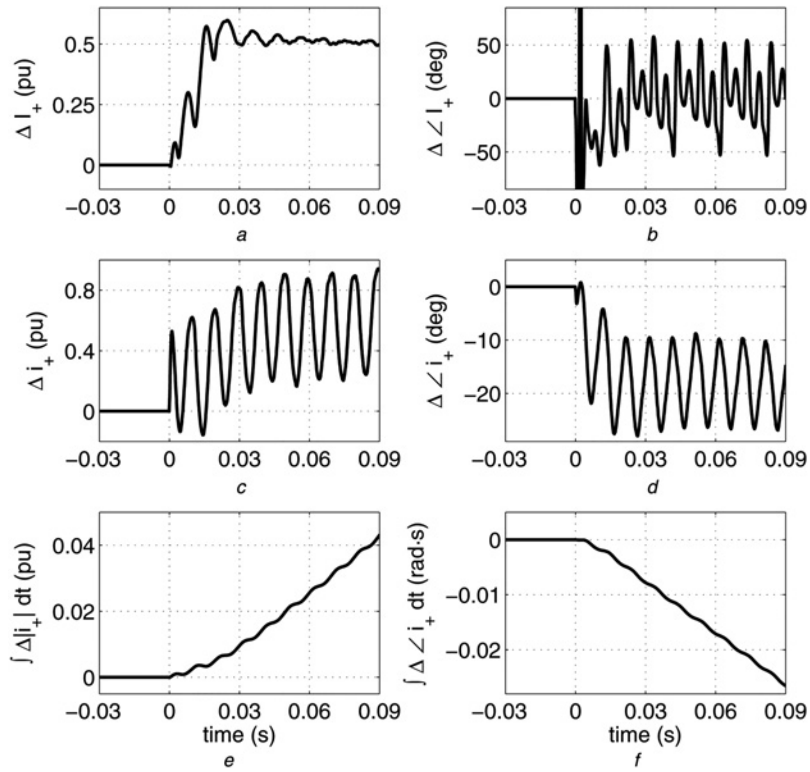

Fig. 8 *Laboratory measurements for a downstream voltage sag (P-P fault at the motor terminals)*

a, *b* Magnitude and phase change (inconclusive results because of the phase change sign variation)
c, *d* Instantaneous magnitude and phase change
e. *f* Integrals for instantaneous magnitude and phase change

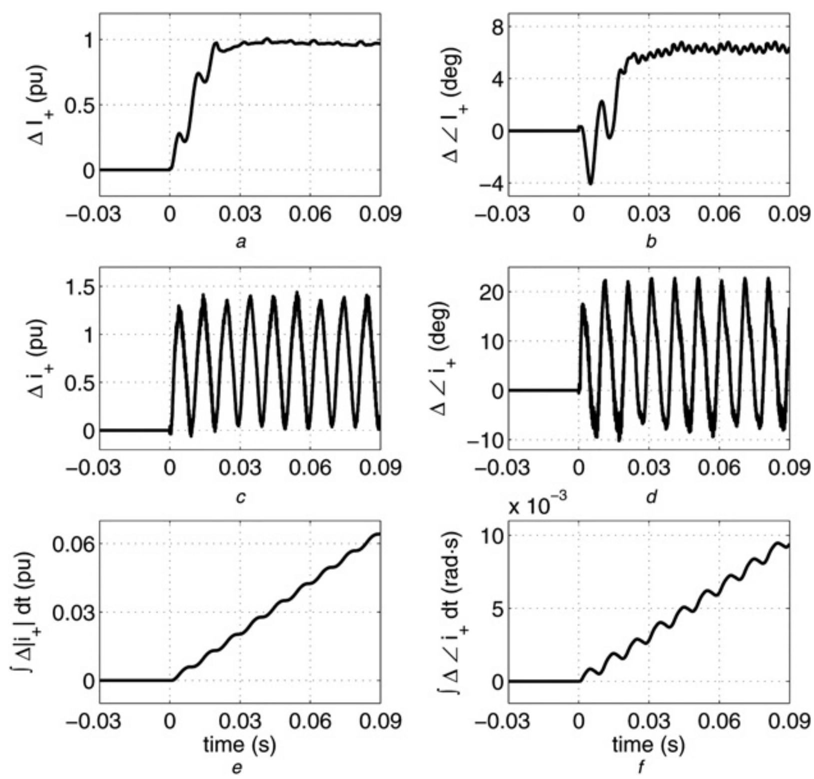

Fig. 9 *Field measurements for a downstream voltage sag (P-E fault in a network with high resistance earthing) where both methods failed as the sign of the phase change was incorrect*

a, *b* Magnitude and phase change
c, *d* Instantaneous magnitude and phase change
e, *f* Integrals for instantaneous magnitude and phase change

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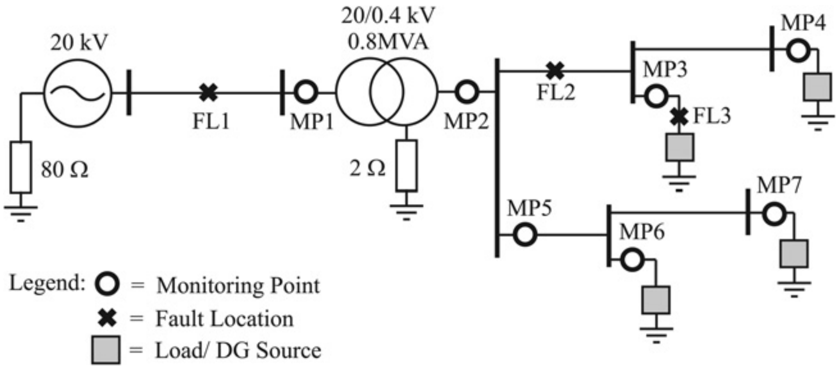

Fig. 10 *Testing radial network for numerical simulations of voltage sags*

3.3 Evaluation of phasor-based and proposed methods

To evaluate the proposed method a similar testing network was used as in [24], as is shown in Fig. 10. Numerical simulations of voltage sags were performed using MATLAB/Simulink, where different combinations of passive inductive loads, induction motors, synchronous motors and constant power loads were used along with DG sources, such as induction and synchronous generators. Voltage sags were generated by 3-P, P-P, P-P-E and P-E faults at three different fault locations (FL1-FL3). Faults with zero and high resistance values were considered, where $R_f > 1\,\Omega$ at the 20 kV level and $R_f > 0.25\,\Omega$ at the 0.4 kV level. Furthermore, along with high resistance earthing (Fig. 10), solid earthing with no earth fault current limitation was used. Voltage sags were also generated with heavy motor starting and loading. Line currents were captured at seven MPs (MP1-MP7) using a sampling frequency of 10 kHz. A simplified dynamic model for a current transformer (CT) was used for capturing line currents. CT magnetic characteristics (mmf against flux) was modelled by piece-wise linear functions, where the magnetic resistance for the saturated region was chosen to be ten-times bigger than for the non-saturated region. Furthermore, hysteresis was not taken into account. All the parameters of the CT model were selected in such a way that the time responses of the IEC class TPZ were obtained [49].

Altogether 507 examples of voltage sags were analysed using numerical simulations, where 209 of them were in the downstream and 298 in the upstream directions. Symmetrical voltage sags were examined separately (133 examples), as well as asymmetrical voltage sags (374 examples) and voltage sags because of earth faults (250 examples). Furthermore, separate analyses were performed for voltage sag events in the testing radial network without DG sources (122 examples for upstream and 206 for downstream directions), where 88 of them were symmetrical and 240 asymmetrical sags (205 examples were because of earth faults). Note that all the testing of examples was performed for zero and high resistance faults, whereas all earth faults were tested for high resistance and solid earthing. Effectiveness was determined for the proposed method (6), (7), as well as for the phasor-based method (5) [30]. The LSQ algorithm was used within the phasor-based method, where only a time window of between 30 ms (1.5 cycle) and 90 ms after the sag inception was observed. In this way any time delay of the LSQ algorithm was excluded. Moreover, for the phasor-based method the cases where either signs of the magnitude or phase changes varied within the observed time window were classified as inconclusive. The obtained effectiveness of the discussed methods is shown in Fig. 11. The results are in complete accordance with laboratory and field testing.

The results presented in Fig. 11*a* show the very high total effectiveness for the proposed method (95%). The effectiveness was reduced to 76%, when only considering the results with the highest degrees of confidence, that is, when the integral criterion (6) and the additional criterion (7) both provided the same result. The proposed method worked almost perfectly for upstream events, whereas an acceptable effectiveness (80–94%) was obtained for downstream events. In contrast, the phasor-based method (5) [30] showed questionable total effectiveness (66%), which was mostly because of the sign variations of the obtained results over time. Furthermore, when events because of downstream high resistance faults were considered the effectiveness was reduced for both methods (Fig. 11*b*), especially within the network without DG sources (Fig. 11*d*). Moreover, both the discussed methods also failed to detect downstream events because of the heavy motor loading. Voltage sags because of heavy motor starting were, on the contrary, all successfully detected by both methods.

Voltage sags because of earth faults were analysed separately as they are the more frequent. The proposed method showed maximal effectiveness for the upstream events (100%), whereas the effectiveness of the phasor-based method was 61% (Fig. 11*a*). On the other hand, the effectiveness for the downstream events was lower, that is, 80% for the proposed method and 50% for the phasor-based method, which was because of the high resistance earthing. When the solid earthing was used then the effectiveness for downstream events was 100% for the proposed method, whereas the effectiveness of the phasor-based method was 70%. In the solidly earthed network without DG sources the effectiveness of both methods was slightly lower for downstream events, that is, 95% for the proposed method and 56% for the phasor-based method.

The main disadvantages for the discussed methods, both the phasor-based and the proposed ones, are the detections of downstream voltage sags because of high resistance faults or because of earth faults in networks with high resistance earthing, as shown in Fig. 11. In such cases the positive-sequence fault current depends mostly on the fault (or earthing) resistance. Moreover, in networks with DG

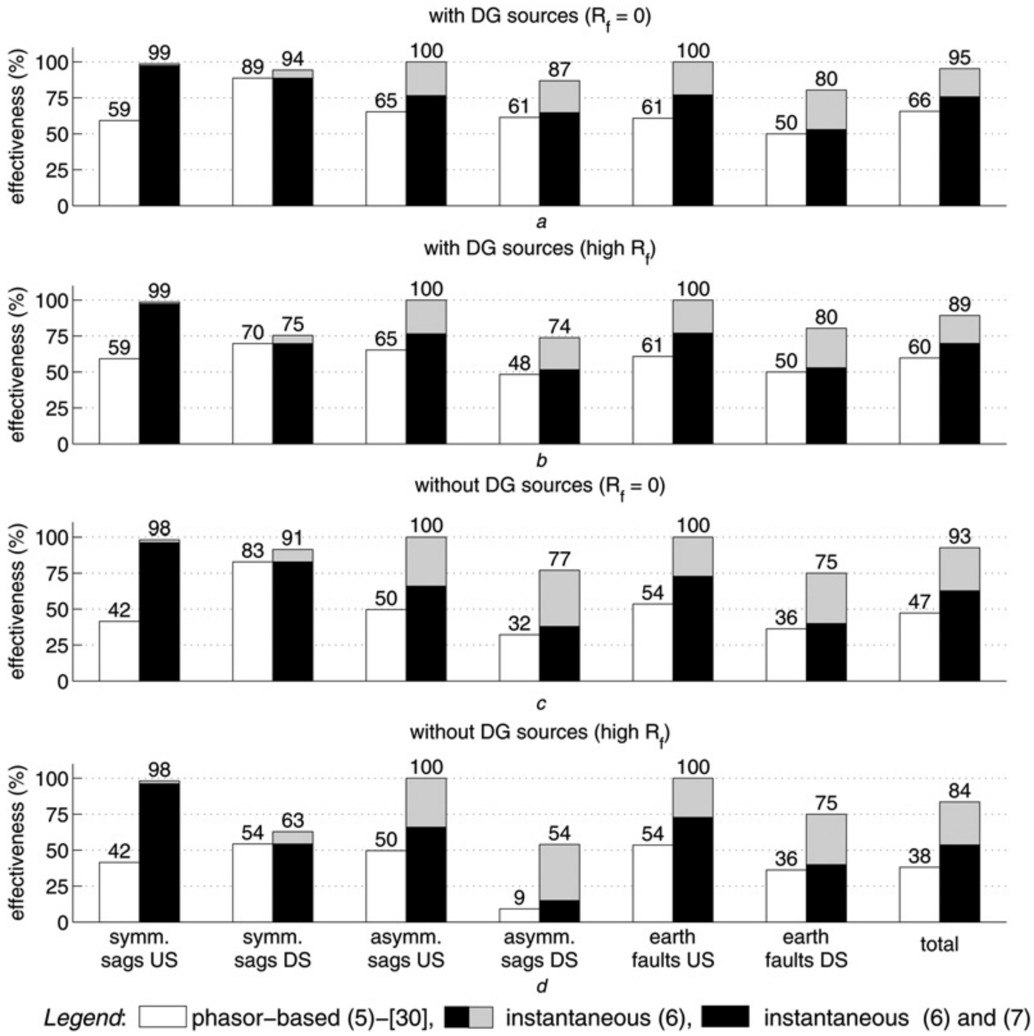

Fig. 11 *Effectiveness of the discussed methods for voltage sag source detection using testing radial network (Fig. 10)*

a With DG sources at $R_f = 0$
b High R_f
c Without DG sources at $R_f = 0$
d High R_f

sources where the supply is at both ends, the phase of the positive-sequence fault current also depends on the pre-fault active and reactive powers [50]. Consequently, the sign of phase change within the positive-sequence current might be incorrect, as is shown in Figs. 9*b*, *d* and *f*. Only the detections of downstream events have deteriorated, as both criteria, the phase and magnitude changes should be fulfilled. Deterioration in detections of downstream events because of the high resistance faults in the testing radial network (Fig. 10) was determined for $R_f > 1\ \Omega$ at the 20 kV level and for $R_f > 0.25\ \Omega$ at the 0.4 kV level. However, the proposed method showed much higher effectiveness than the phasor-based method.

Furthermore, the impact of CT saturation was determined. The effectiveness was unaffected for the proposed method, whereas the results obtained by the phasor-based method were significantly delayed because of severe signal distortion.

4 Conclusion

This paper proposed a current-based method for detecting voltage sag sources that is derived in the time domain. Instantaneous symmetrical components are applied, whereas only the instantaneous positive-sequence current is used as a criterion. The introduced approach generalises methods known from the literature that use the conventional phasor estimation techniques. Compared with the phasor-based methods, the proposed method is generally faster, more effective and shows a higher degree of confidence, especially for transient voltage sag events in networks with DG and active loads. Furthermore, implementation of the proposed method on powerful signal processors, built-in modern numerical directional overcurrent relays, is not as significantly demanding with respect to the already implemented methods.

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